7 **(a)** Fig. 7.1 shows three bars of steel, A, B and C.

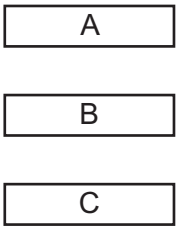

Fig. 7.1

A student is given the three pieces of steel. Two of the pieces are magnetised and one piece is unmagnetised.

Describe and explain how the student determines which piece is unmagnetised using only the three pieces of steel.

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..... [4]

(b) Fig. 7.2 shows a circuit diagram of a step-down transformer.

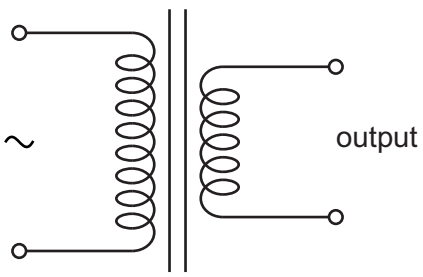

Fig. 7.2

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(i) The mains voltage supplied to the transformer is 240 V. The output power of the transformer is 45 W. The transformer is 100% efficient.

Calculate the input current to the transformer.

input current = ... [3]

(ii) Draw a labelled diagram of a step-down transformer. On the labels, state a suitable material for each of the components.

[3]

[Total: 10]

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